Statistics for Computer Scientists

Lecture 2: Descriptive statistics

12 September 2025

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The Netherlands

Where are we

Lecture	Topic	Book
5 Sept	Introduction, statistics fundamentals	Chapter 1
12 Sept	Descriptive statistics	Chapter 2
19 Sept	Bivariate relationships	Chapter 3
26 Sept	Data collection	Chapter 4
10 Oct	Sampling distributions	Chapter 7
17 Oct	Estimation	Chapter 8
31 Oct	Significance testing	Chapter 9
7 Nov	Comparing two groups	Chapter 10
14 Nov	Categorical data analysis	Chapter 11
21 Nov	Linear regression	Chapter 12
28 Nov	Multiple regression	Chapter 13
5 Dec	Tba	
12 Dec	Tba	

Today's lecture

- Recap lecture 1

- Variable types
  - Categorical, quantitative
- Graphical data description
- Numerical data description

Recap

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Recap

- Importance of statistics

- Sample vs population
- Descriptive and inferential statistics
- Variability: **within** and **between** sample variation
- Randomness and margin of error

Check your understanding

Biologists want to estimate the mean body mass for penguins. They measure body mass in 450 penguins and find a mean of 3.2 kg. Based on this, they estimate that the mean body mass of all penguins is between 2.8 – 3.7 kg.

1. What is the sample, what is the population?
2. Which part is descriptive, which part is inferential?
3. What is the statistic, and what is the parameter?
4. What is the variable of interest?
5. What type of variable is this?

Variable types

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Mean and proportion

- How tall are university students in the Netherlands?
  - Parameter: the average height of all university students in NL (population mean)
  - Statistic: the average height of all students in this class (sample mean)

- How popular is a certain political party among people >65 years old?
  - Parameter: the percentage of all people over 65 years old that voted for this party (population proportion)
  - Statistic: the percentage of 250 inhabitants in a particular elderly home that voted for this party (sample proportion)

Categorical and quantitative variables

Categorical variables:

- Favorite color
- Pass/fail a course
- Likert scales*

The values of a categorical variable are restricted to several **categories**

(e.g., strongly agree – agree – neutral – disagree – strongly disagree in a survey)

Quantitative variables:

- Cups of coffee this morning
- Height
- Travel time to campus

The values of a quantitative variable are **numerical values** (that represent different magnitudes of the variable!)

penguins (C:\Users\mvink\OneDrive\Teaching\STAT2425\Datasets)

Edit Data | Descriptives | T-Tests | ANOVA | Mixed Models | Regression | Frequencies

	species	island	beak_length	beak_depth	flipper_length	body_mass	sex
1	Adelie	Torgersen	39.1	18.7	181	3750	male
2	Adelie	Torgersen	39.5	17.4	186	3800	female
3	Adelie	Torgersen	40.3	18	195	3250	female
4	Adelie	Torgersen	39	19.1	194	3300	female
5	Adelie	Torgersen	36.7	19.3	193	3450	female
6	Adelie	Torgersen	39.3	20.6	190	3650	male
7	Adelie	Torgersen	38.9	17.8	181	3625	female
8	Adelie	Torgersen	39.2	19.6	195	4675	male

Categorical: nominal and ordinal

- Is there an ordering in the categories?

- Nominal: no ordering
  - Favorite color, pass/fail, gender, music genre

- Ordinal: order exists
  - Likert scale, letter grading system, binned variables (baby, child, adolescent, adult)
  - Can sometimes be treated as a quantitative variable

Quantitative: discrete and continuous

- Which (numeric) values can the variable take?

- Discrete: limited set of possible values
  - Cups of coffee this morning
  - Often “the number of…” -> only integers

- Continuous: possible values are an interval (in theory unlimited values)
  - Height, travel time to campus

Mean and proportion

Lecture 1 slide 34

- How tall are university students in the Netherlands?
  - Parameter: the average height of all university students in NL (population mean)
  - Statistic: the average height of all students in this class (sample mean)

- How popular is a certain political party among people >65 years old?
  - Parameter: the percentage of all people over 65 years old that voted for this party (population proportion)
  - Statistic: the percentage of 250 inhabitants in a particular elderly home that voted for this party (sample proportion)

Graphical data descriptions

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Summarising data (samples)

- Frequency table: list of all possible values and their frequency in the sample

- Categorical: frequency of each value
- Quantitative: frequency of value intervals

Frequency table – categorical

- Example: variable *species* from penguins.csv

Frequencies for species

species	Frequency	Percent
Adelie	152	44.186
Chinstrap	68	19.767
Gentoo	124	36.047
Missing	0	0.000
Total	344	100.000

Adelie

Chinstrap

Gentoo

Frequency table – quantitative

- Example: variable *body_mass* from penguins.csv
- Too many unique values, use intervals for table!

body_mass	Frequency
2700 – 3000	9
3000 – 3300	25
3300 – 3600	27
etc	

Frequencies for body_mass ▼

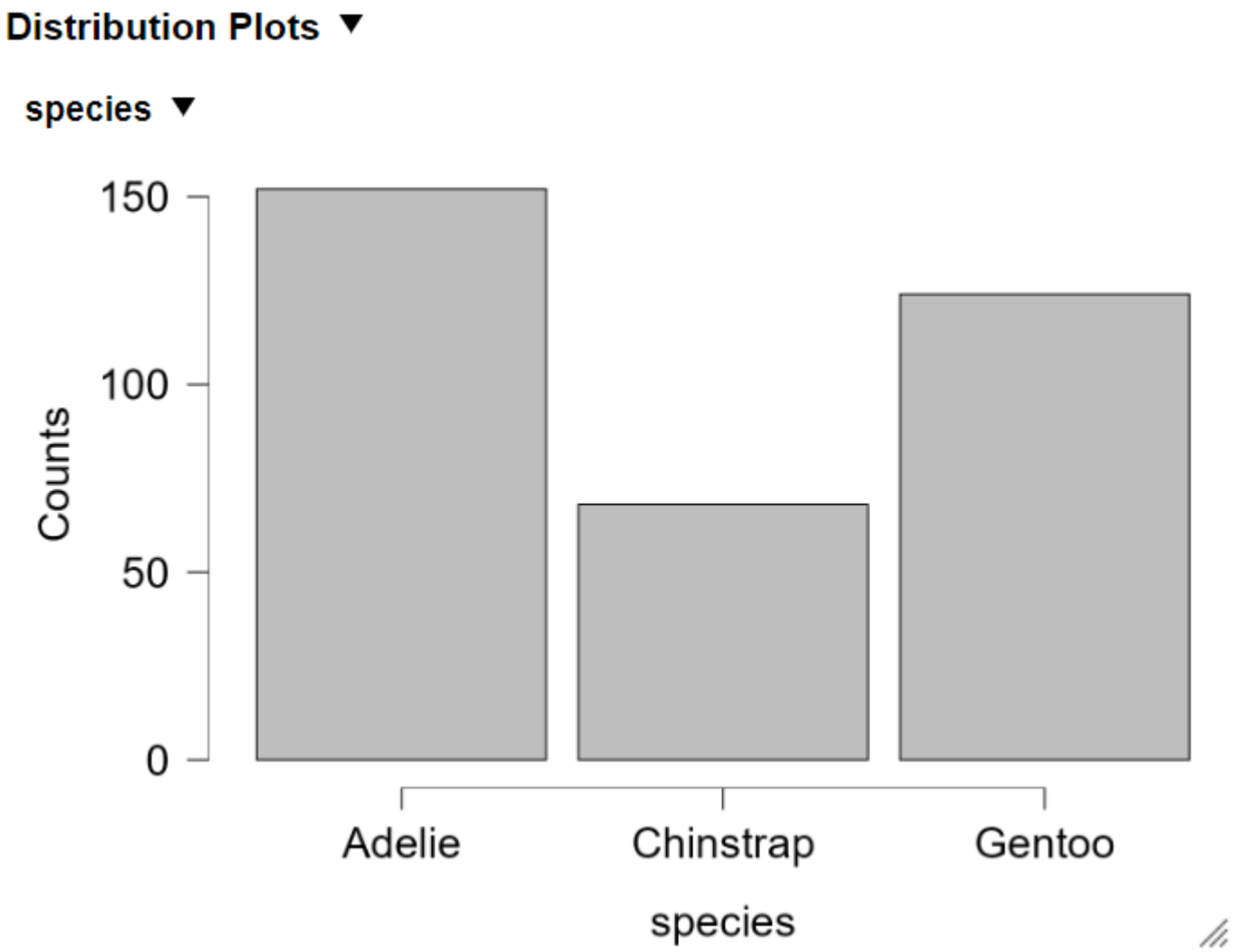
body_mass	Frequency	Percent
2700	1	0.291
2850	2	0.581
2900	4	1.163
2925	1	0.291
2975	1	0.291
3000	2	0.581
3050	4	1.163
3075	1	0.291
3100	1	0.291
3150	4	1.163
3175	2	0.581
3200	5	1.453
3250	5	1.453
3275	1	0.291
3300	7	2.035
3325	5	1.453
3350	5	1.453
3400	8	2.326
3425	2	0.581

Bar plot for categorical data

- Categorical data: bar plot
- Pareto: ordered by frequency

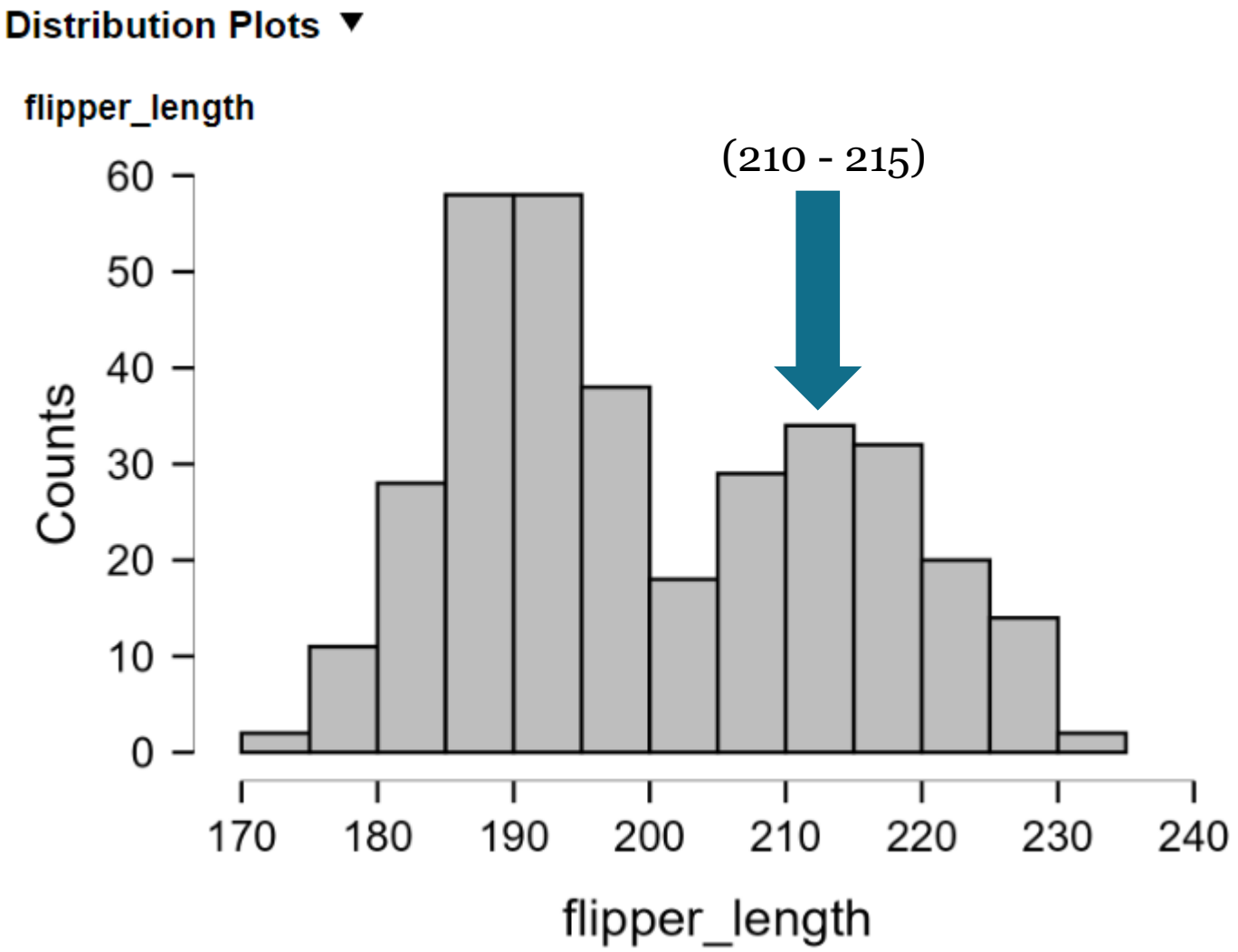
Frequencies for species

species	Frequency
Adelie	152
Chinstrap	68
Gentoo	124
Missing	0
Total	344



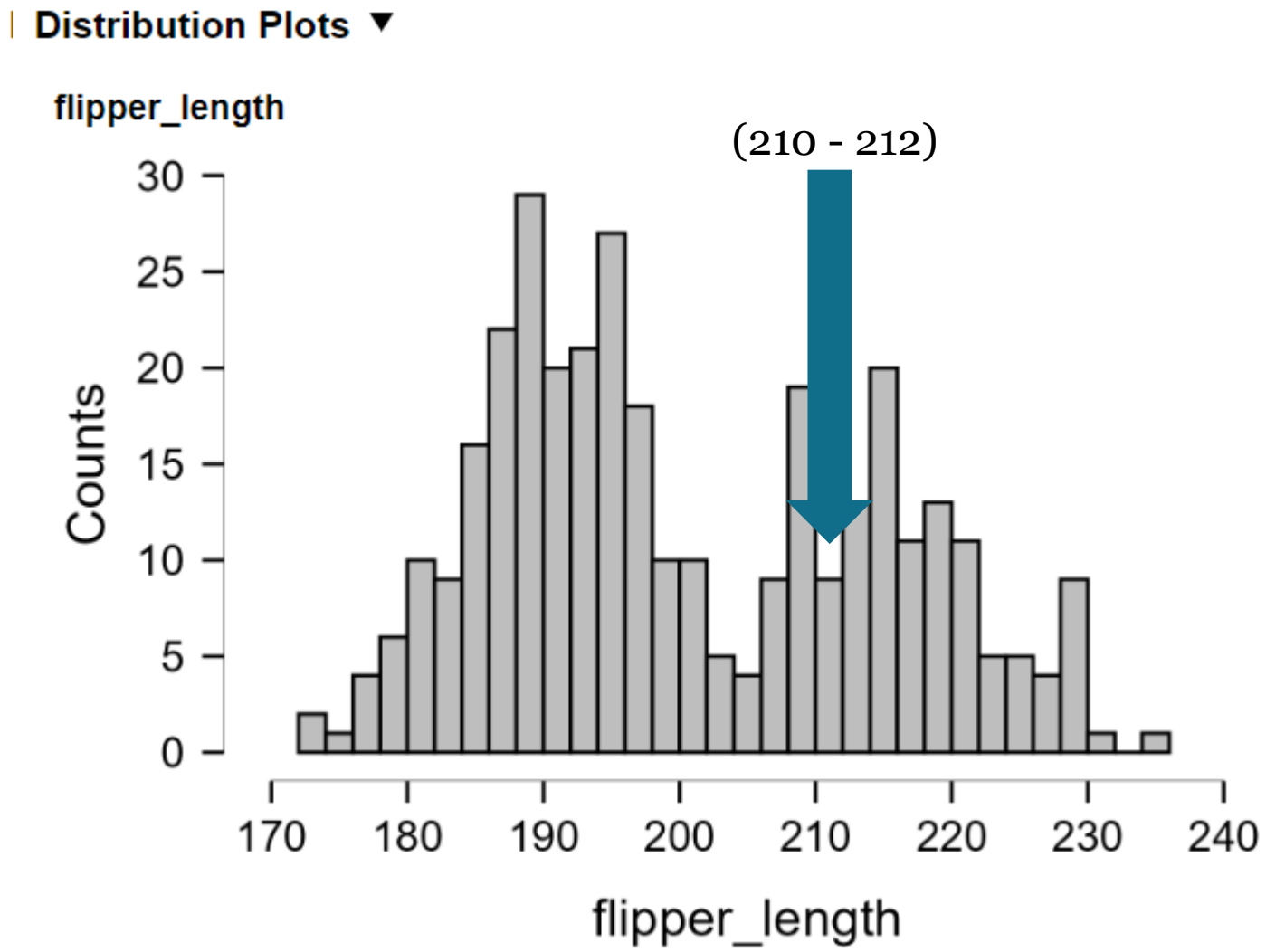
Histogram for continuous data

flipper_length
181
186
195
194
193
190
181
195
193

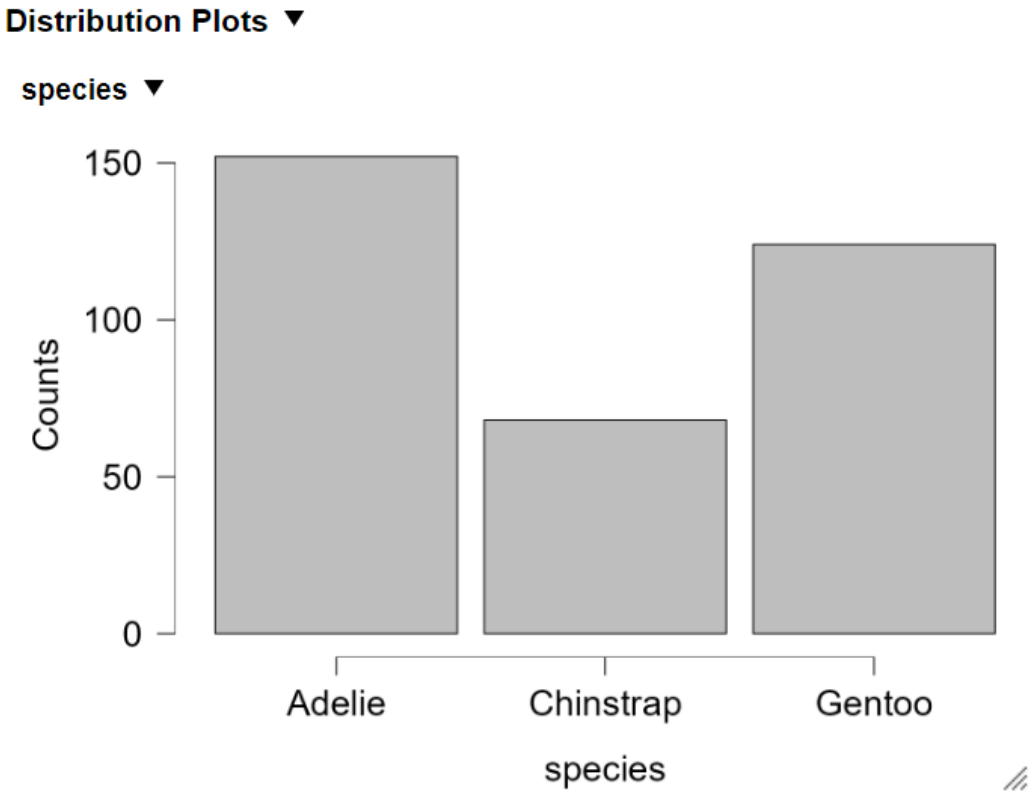


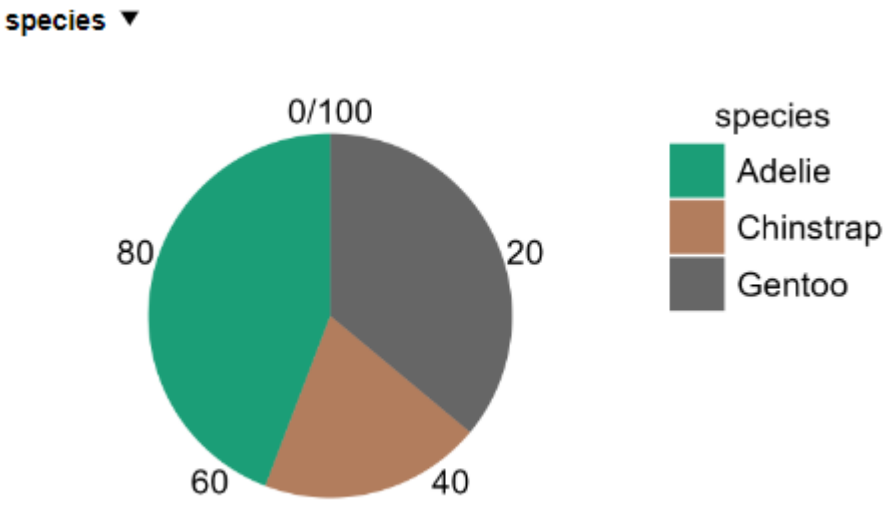
Histogram for continuous data

flipper_length
181
186
195
194
193
190
181
195
193



Side note: pie charts





Guidelines

Pie chart:

- Part-to-whole comparison
- Small number of categories (max ~5)

Bar chart:

- Group-group comparison
- Any number of categories
- Safest choice

Please don’t ☹

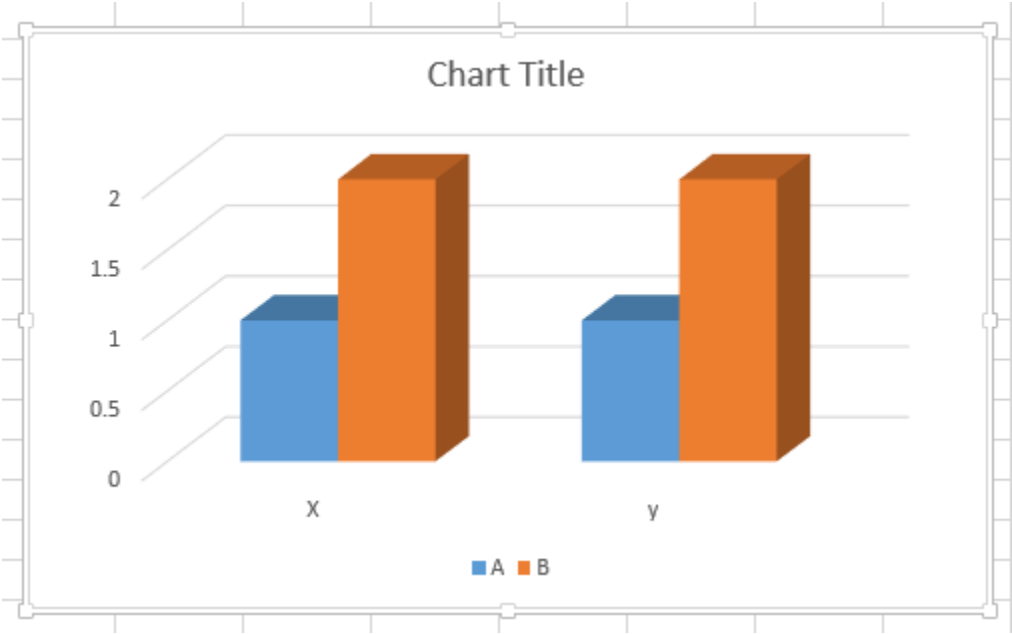
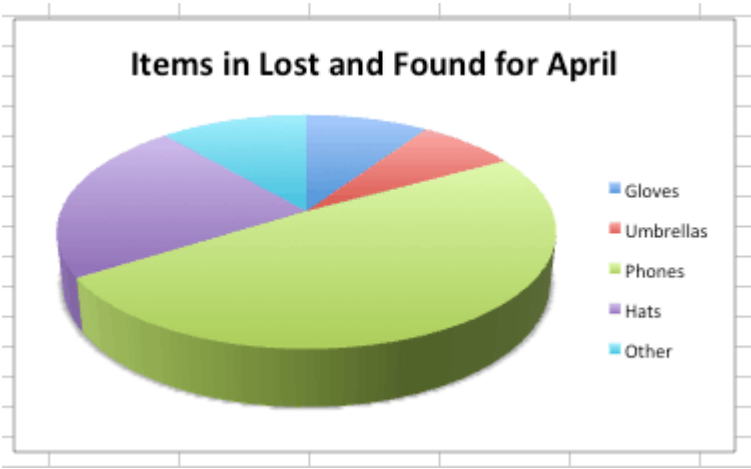
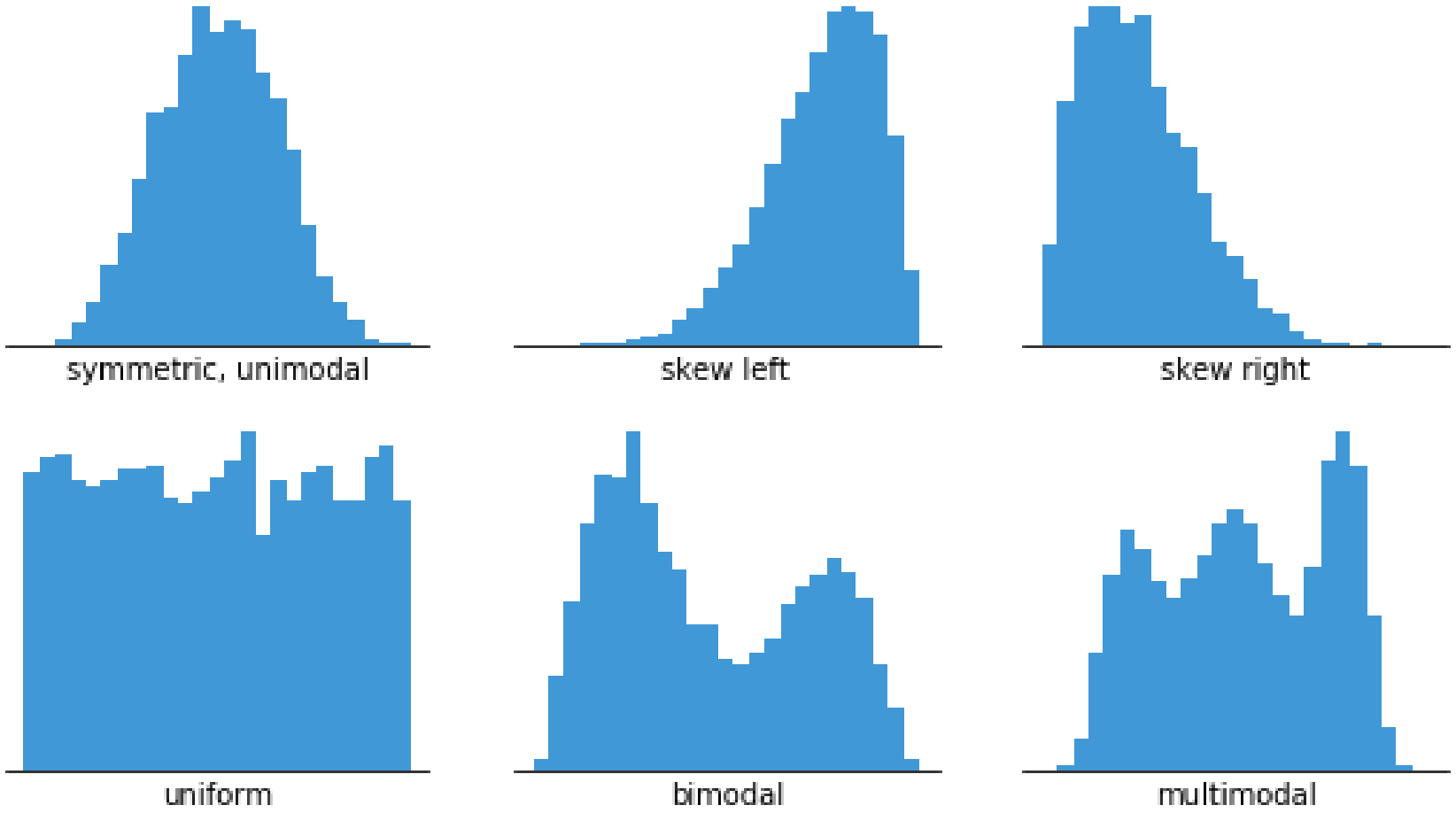


Chart Title
2
1.5
1
0.5
0
X
y
A
B


Items in Lost and Found for April
Gloves
Umbrellas
Phones
Hats
Other

The many shapes of histograms



Histogram shapes

- Bell-shaped/symmetric/normal (height, IQ)
- Skewed left (score on an easy exam)
- Skewed right (annual income, number of arrests)
- Uniform (fair die)
- Bimodal (polarized opinions)

Numerical descriptions

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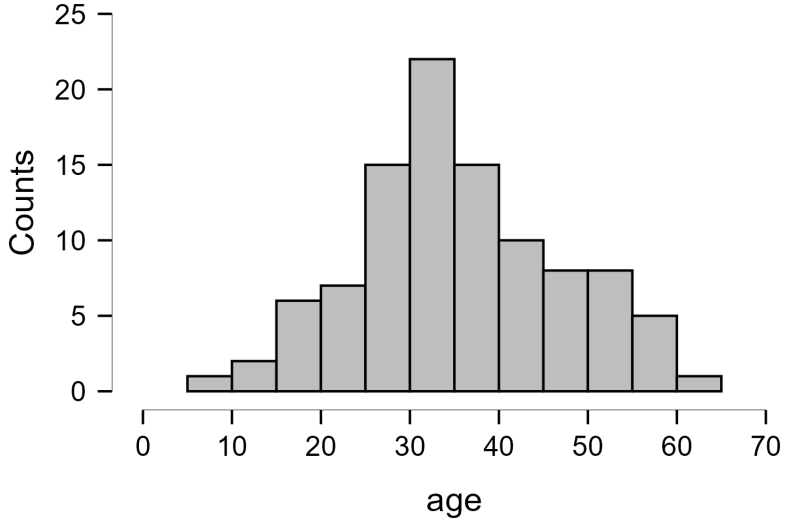
Describing the center

- Let Y denote a quantitative variable with observations $y_1, y_2, y_3, \ldots, y_n$

- Describing the **center**
- **Mean:** sum all, divide by n: $\bar{y} = \frac{y_1 + y_2 + \cdots + y_n}{n} = \frac{\sum y_i}{n}$
- **Median**: middle value in ordered sample
- **Mode:** most frequent value



Example: annual CO2 emission

- Mean, median and mode?

Country	Emission
Bangladesh	0.3
Brazil	1.8
China	2.3
India	1.2
Indonesia	1.4
Pakistan	0.7
Russia	9.9
United States	20.1

Example: annual CO2 emission

- Mean, median and mode?

- $mean$: $\bar{y} = {\Sigma y_i}/{n} = {37.7}/{8} = 4.7$

Country	Emission
Bangladesh	0.3
Brazil	1.8
China	2.3
India	1.2
Indonesia	1.4
Pakistan	0.7
Russia	9.9
United States	20.1

Example: annual CO2 emission

- Mean, median and mode?

- Median: first order the sample
- Then take the middle value(s)
- $Median = \frac{1.4+1.8}{2} = 1.6$

Country	Emission
Bangladesh	0.3
Pakistan	0.7
India	1.2
Indonesia	1.4
Brazil	1.8
China	2.3
Russia	9.9
United States	20.1

Example: annual CO2 emission

- What about the mode?

- There isn't one (each value occurs once)
- Mode usually not as useful for quantitative variables (unless binned)

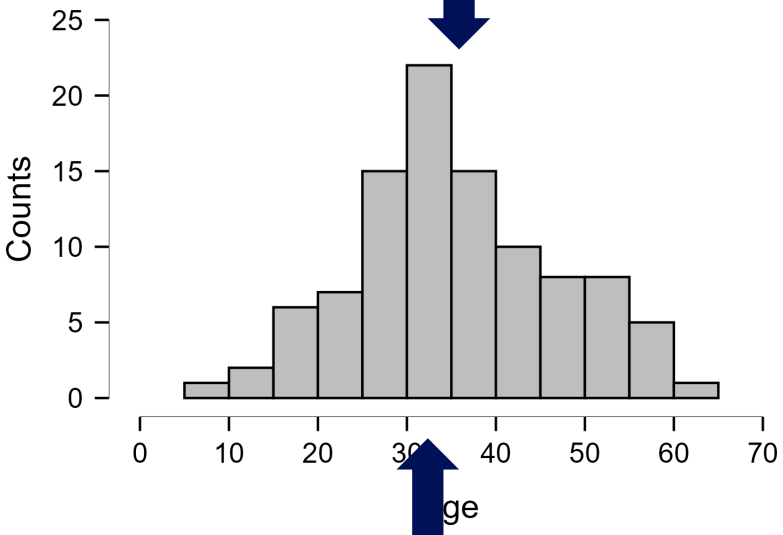
Country	Emission
Bangladesh	0.3
Brazil	1.8
China	2.3
India	1.2
Indonesia	1.4
Pakistan	0.7
Russia	9.9
United States	20.1

Properties of mean and median

- For symmetric distributions: mean = median
- For skewed distributions, mean is drawn in the direction of the tail
- Mean: valid for interval and ratio scales
- Median: valid for interval, ratio and ordinal scales

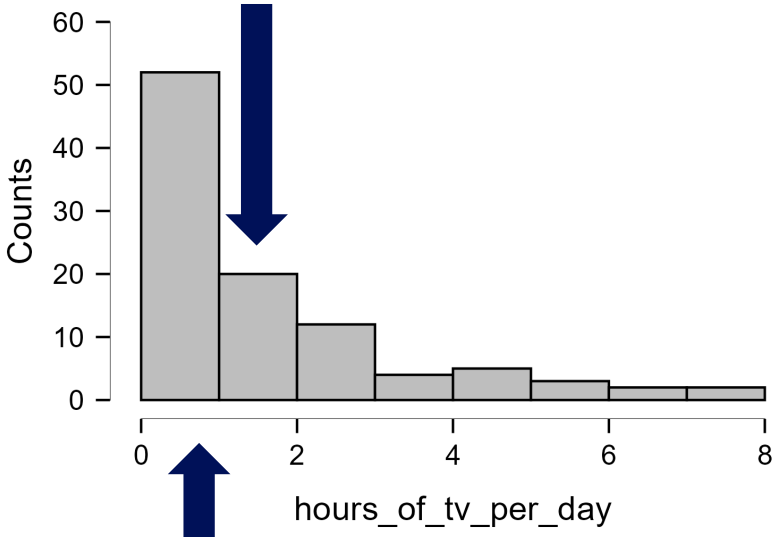
- Mean is more sensitive to outliers
- Slightly skewed distribution: mean is preferred
- Highly skewed distribution: median is preferred

mean: 35



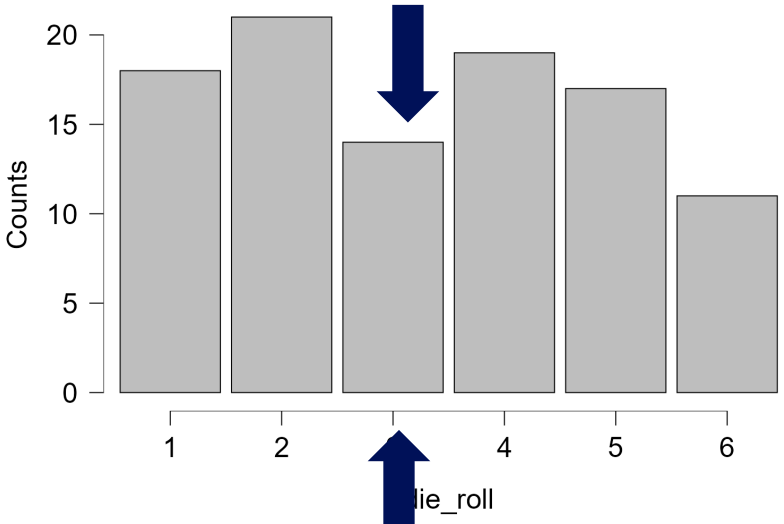
median: 34.9

mean: 1.6



median: 0.9

mean: 3.3



median: 3

Modes?

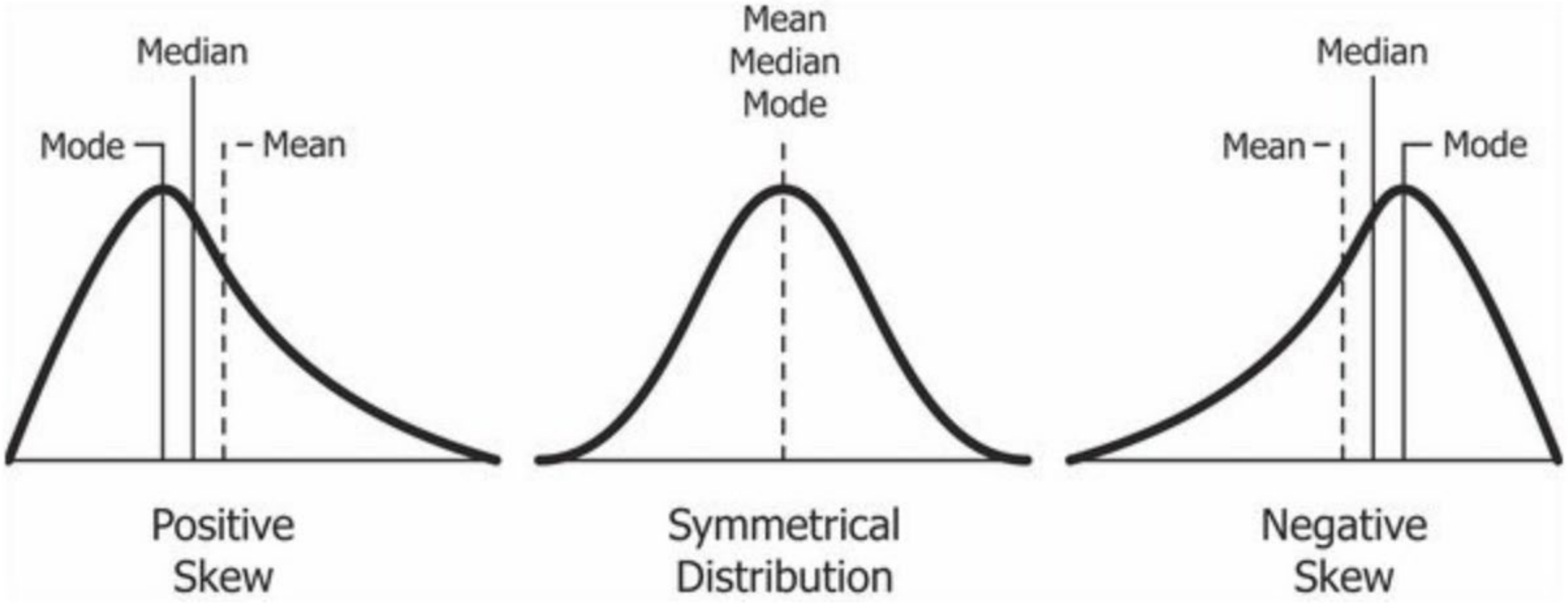
Sensitivity to outliers

Country	Emission
Bangladesh	0.3
Pakistan	0.7
India	1.2
Indonesia	1.4
Brazil	1.8
China	2.3
Russia	9.9
United States	20.1

- Mean: 4.7
- Median: 1.6

Country	Emission
Bangladesh	0.3
Pakistan	0.7
India	1.2
Indonesia	1.4
Brazil	1.8
China	2.3
Russia	9.9
United States	200000.1

- Mean: 250002.2
- Median: 1.6



Right-skewed No skew Left-skewed

Describing categorical variables

- Proportion: the ratio of the number of occurrences of a set of categories to the total number of values

- For a sample, this is denoted by $\hat{\pi}$ and given by:

$$\hat{\pi} = \frac{o}{n},$$

- where *o* is the number of times the given category occurs and *n* is the sample size

Proportions

Frequency Tables ▼

Frequencies for species

species	Frequency	Percent	Valid Percent	Cumulative Percent
Adelie	151	43.895	44.023	44.023
Chinstrap	68	19.767	19.825	63.848
Gentoo	124	36.047	36.152	100.000
Missing	1	0.291		
Total	344	100.000		

- $\hat{\pi}(Adelie) = \frac{151}{344} = 0.44$

$\hat{\pi}(Chinstrap) = \frac{68}{344} = 0.20$

Describing variability

- Range
  - Difference between largest and smallest observations
  - Range $= 20.1 - 0.3 = 19.8$

- Standard deviation
  - The "typical" distance from the mean
  - Denoted with s, it is the square root of the variance
  - Variance: $s^2 = \frac{\sum(y_i - \bar{y})^2}{n-1}$
  - Standard deviation: $s = \sqrt{\frac{\sum(y_i - \bar{y})^2}{n-1}}$

Country	Emission
Bangladesh	0.3
Pakistan	0.7
India	1.2
Indonesia	1.4
Brazil	1.8
China	2.3
Russia	9.9
United States	20.1

Calculating standard deviation

1. Calculate the mean
2. For every observation, calculate difference from mean
3. Square those differences
4. Add them up
5. Divide by (n-1) -> variance
6. Take square root -> standard deviation

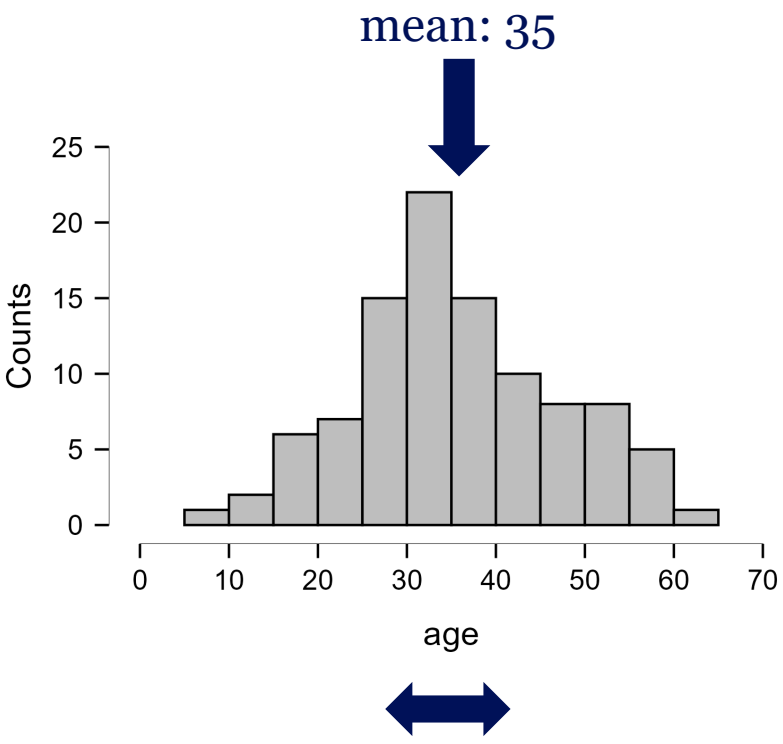
Family	Number of children
Smith	1
Jones	3
Brown	2
Williams	6
Patel	4
Stevens	2

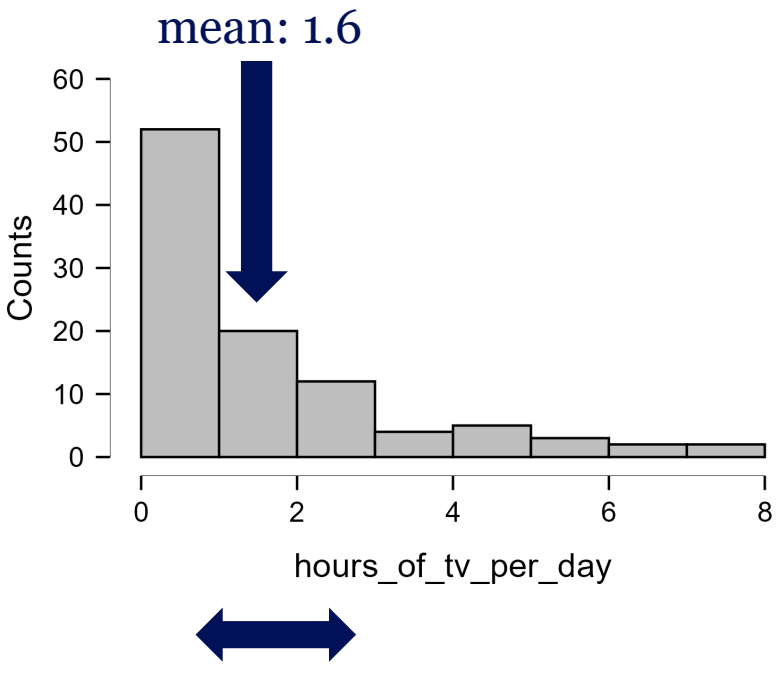
Calculating standard deviation

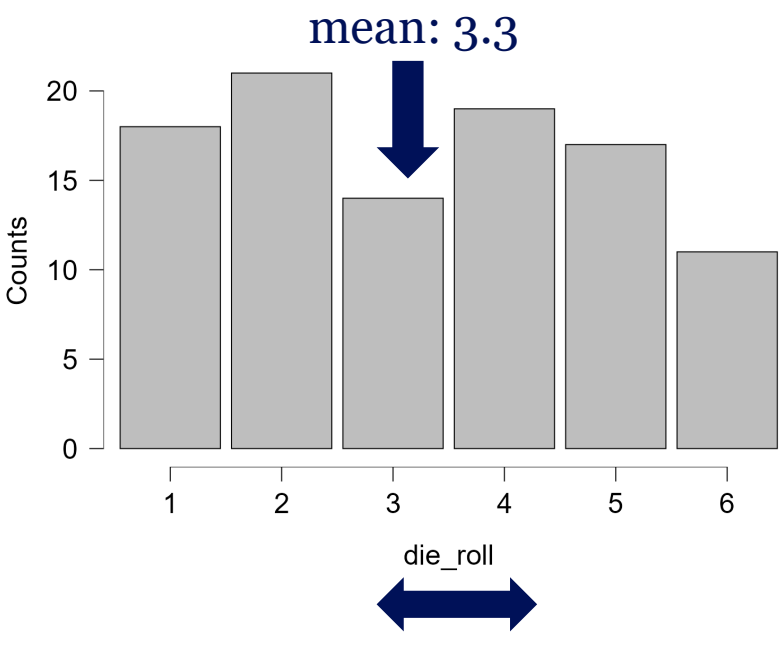
1. $\bar{y} = \frac{1+3+2+6+4+2}{6} = \frac{18}{6} = 3$
2. See third column
3. See fourth column
4. $4 + 0 + 1 + 9 + 1 + 1 = 16$
5. $s^2 = \frac{16}{6-1} = 3.2$
6. $s = \sqrt{3.2} = 1.8$

Family	Number of children (y)	$(y - \bar{y})$	$(y - \bar{y})^2$
Smith	1	-2	4
Jones	3	0	0
Brown	2	-1	1
Williams	6	3	9
Patel	4	1	1
Stevens	2	-1	1

Note: unit of s is equal to unit of variable (here: children) -> easy interpretation


mean: 35
Counts
25
20
15
10
5
0
0
10
20
30
40
50
60
70
age
$s = 11$


mean: 1.6
Counts
60
50
40
30
20
10
0
0
2
4
6
8
hours_of_tv_per_day
$s = 1.7$

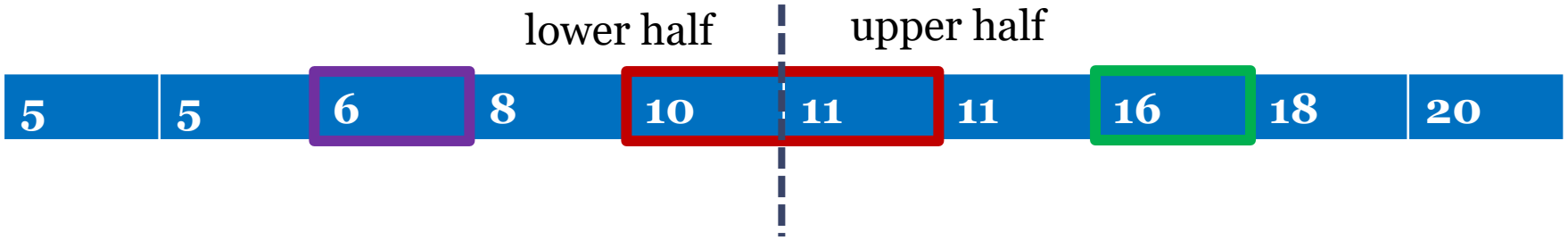

mean: 3.3
Counts
20
15
10
5
0
1
2
3
4
5
6
die_roll
$s = 1.7$

Measures of position

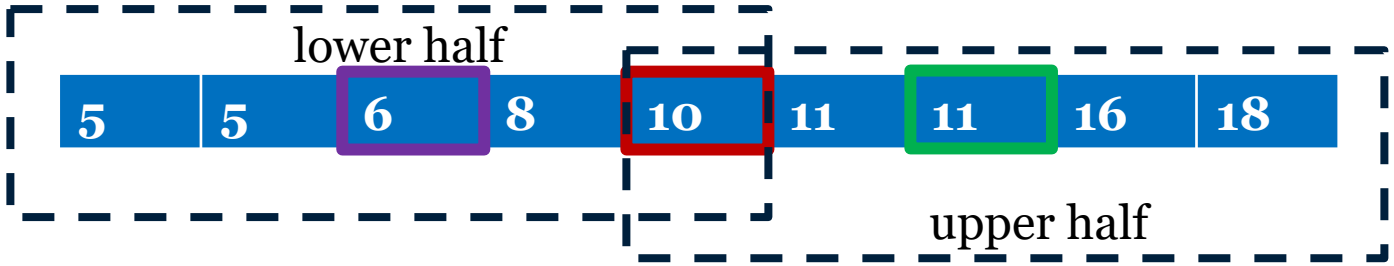
- Percentile: p% of observations are below the pth percentile
- 25th percentile = first quartile, Q1
- 50th percentile = median, also called 2nd quartile
- 75th percentile = third quartile, Q3

- Interquartile range: $IQR = Q3 - Q1$ or given as $[Q1, Q3]$

Quartiles



Median (Q2): 10.5
First quartile (Q1): 6
Third quartile (Q3): 16



Median (Q2): 10
First quartile (Q1): 6
Third quartile (Q3): 11

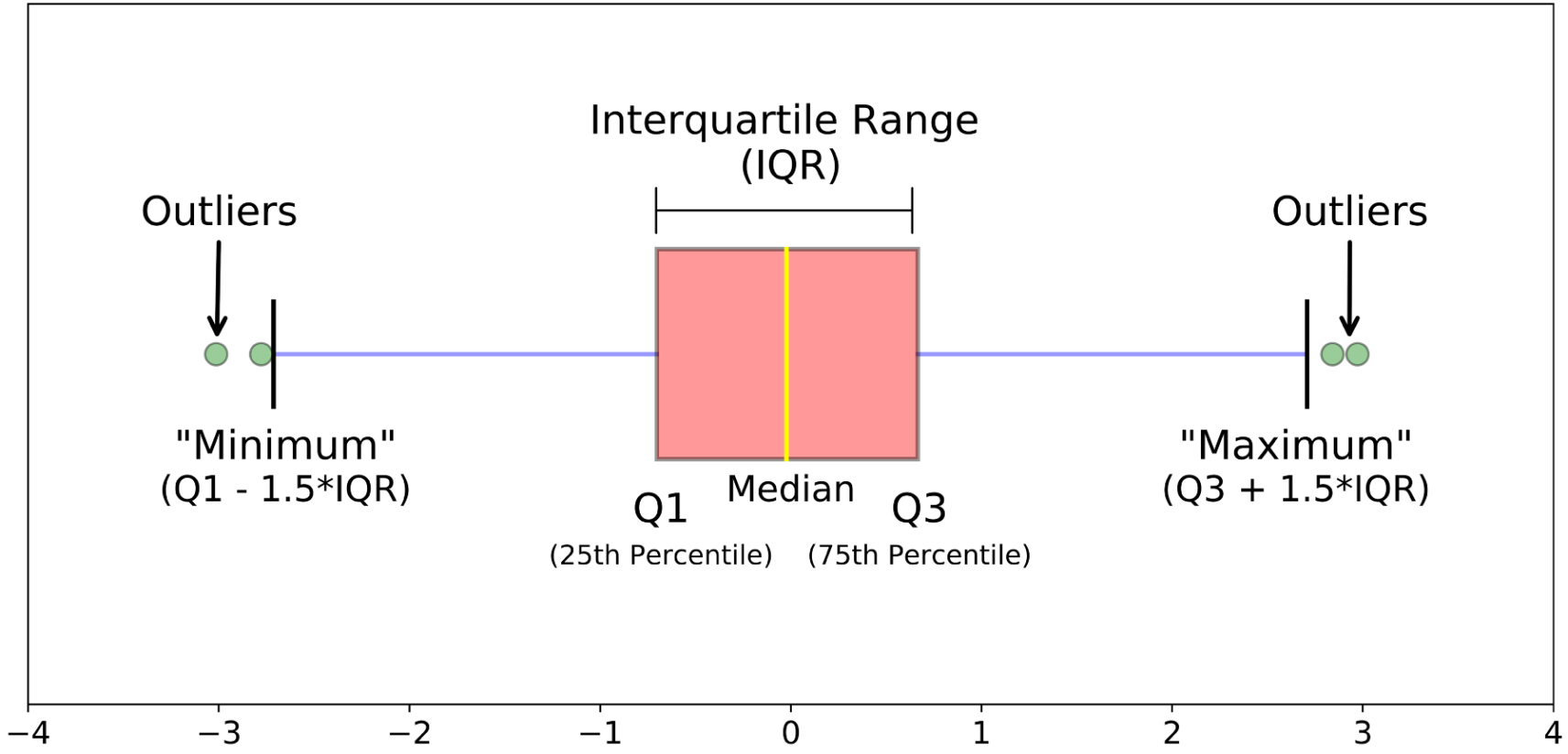
Note on notation (important!)

- Mean, proportion, variance and standard deviation exist for both the sample and the population
- Sample: calculated, notation with Latin characters
- Population: usually estimated, notation with Greek characters
  - (Except proportion! Because we use p for probability…)

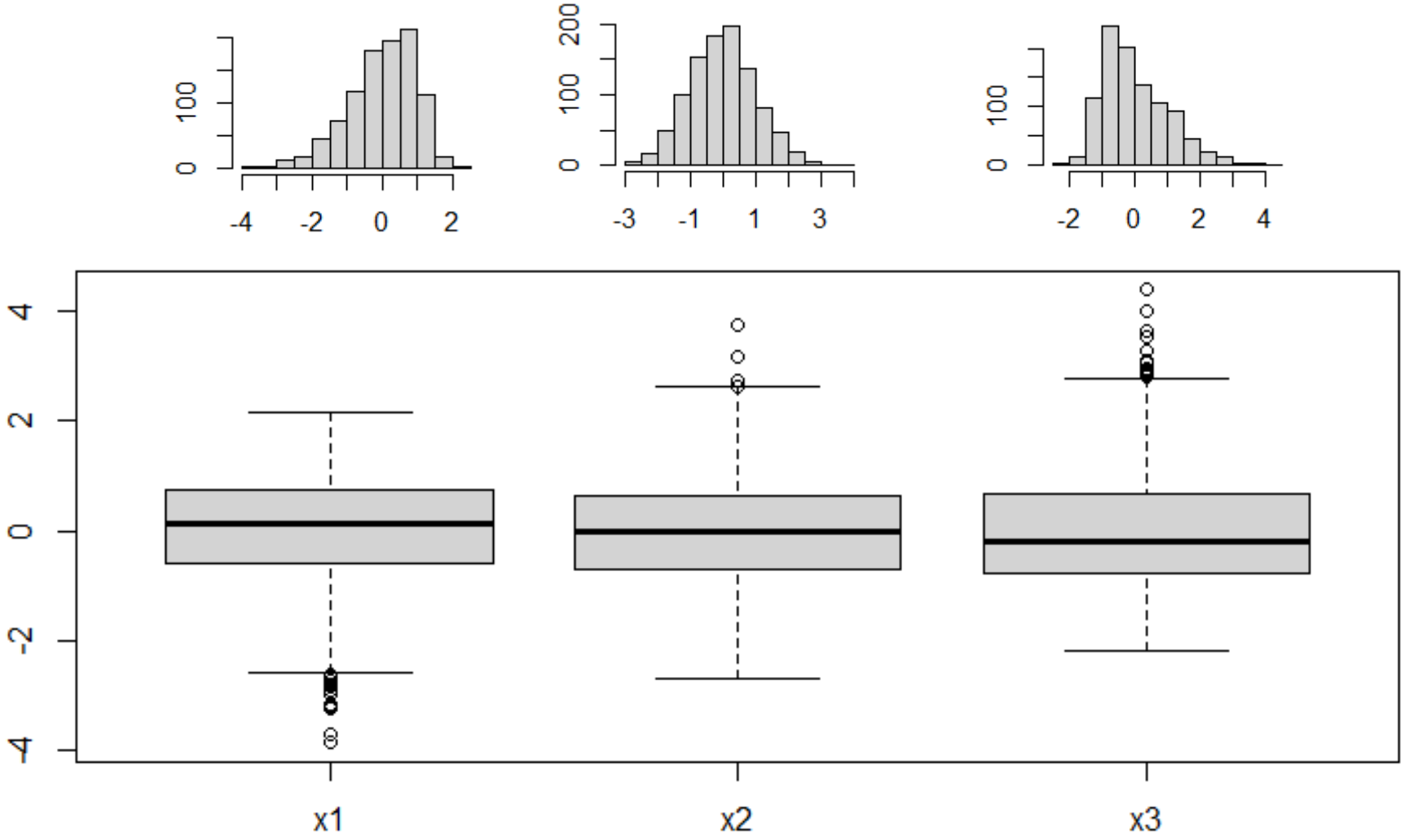
	Sample (statistic)	Population (parameter)
Mean	$\bar{y}$	μ
Proportion	$\hat{\pi}$	π
Variance	s^2	σ^2
Standard deviation	s	σ

Another graph: box plot

- Shows various numeric statistics in a single plot



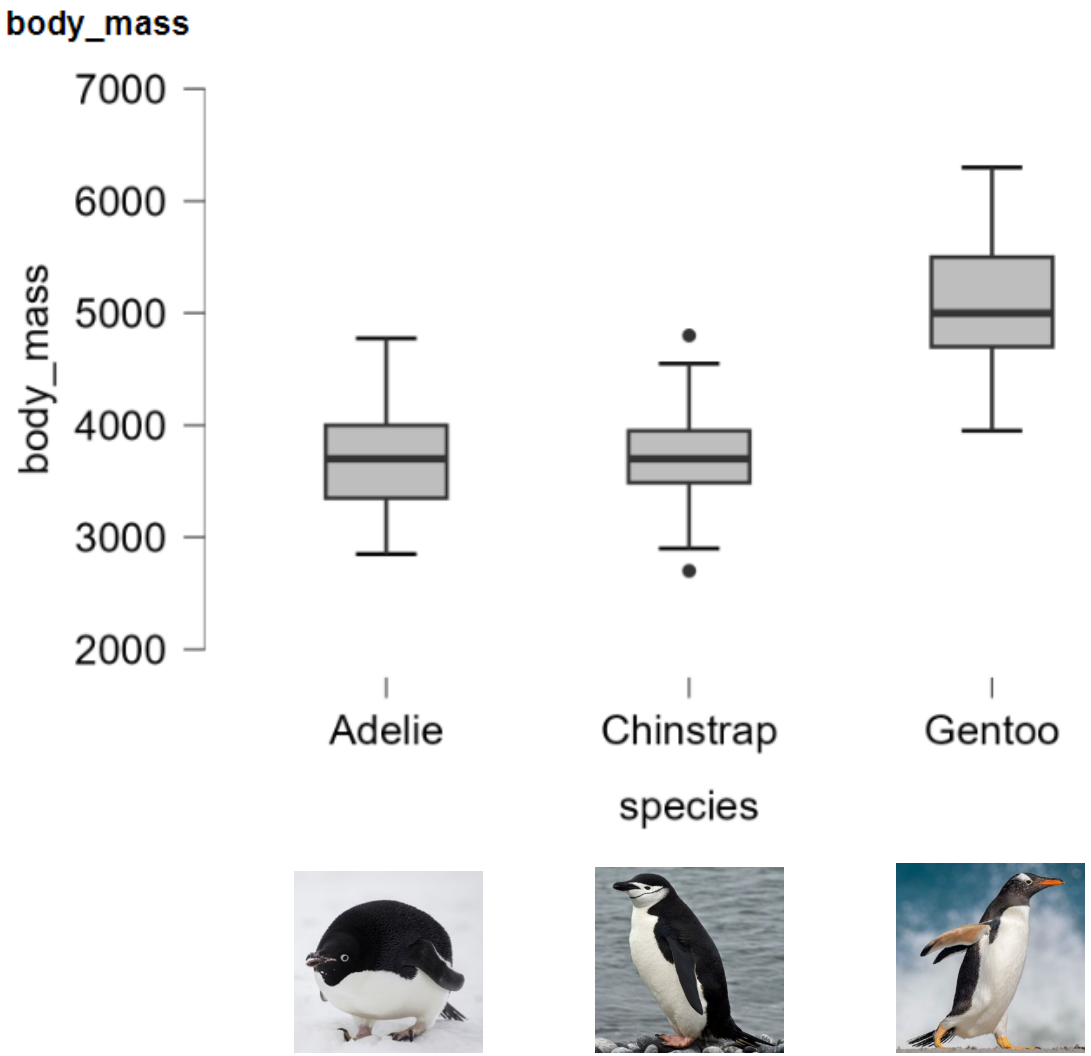
Box plot vs histogram



Side-by-side box plots

- One quantitative and one categorical variable: make a box plot for each category and show them side-by-side

- Penguins example: body mass by species



Linear transformations and standardizing

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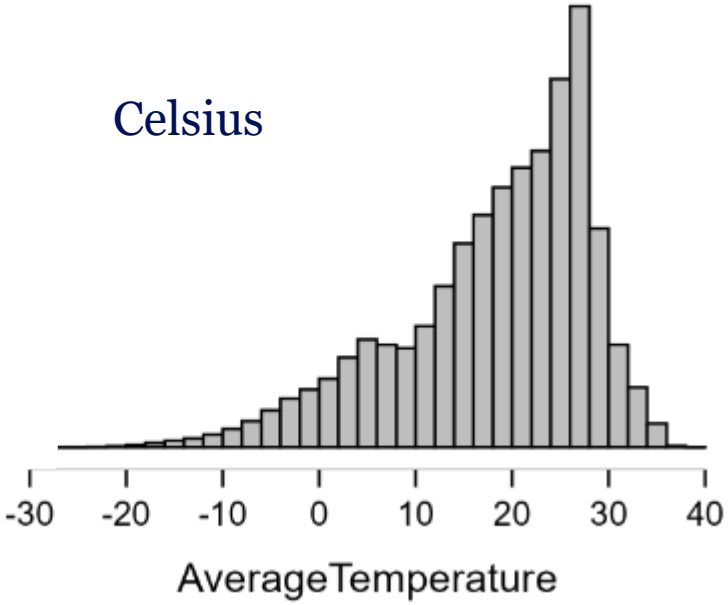
What is a linear transformation?

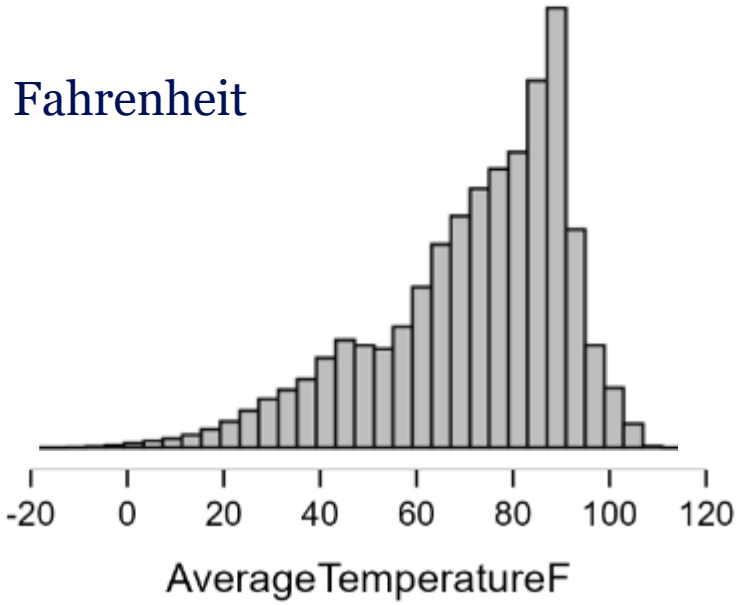
- Any change to a variable using only scaling and shifting
  - Scaling: multiplying/dividing by a constant
  - Shifting: adding/subtracting a constant

- Examples:
  - $bodymass_{gram} = bodymass_{kg} * 1000$ (no shift, only scale)
  - $temp_{celsius} = \left(temp_{fahrenheit} - 32\right) * \frac{5}{7}$ (shift and scale)

Effects on distribution

- Shape of distribution **does not change**!
- Mean and median change with same formula
- Standard deviation is only affected by scaling, not shifting





Standardizing

- A special form of linear transformation
  - Shift by subtracting the mean from all observations
  - Scale by dividing all observations by the standard deviation

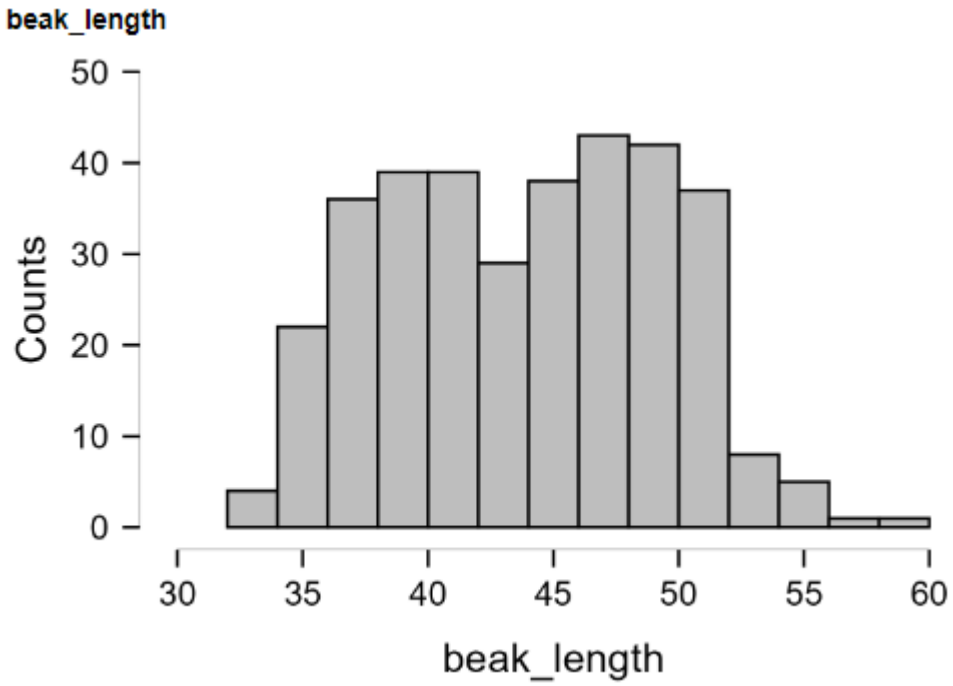
- Result: mean of 0 and sd of 1!
- Also called z-scores
- Intuitively: equivalent value of **standard normal distribution**

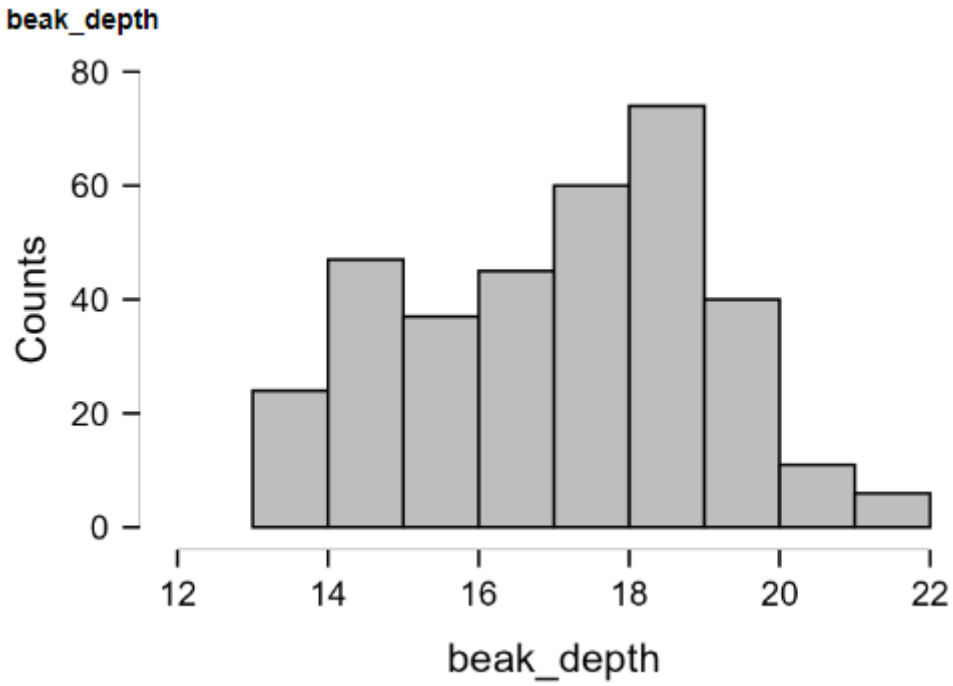
Comparing measures of position

- Which is more extreme, a value of 30 in N(25, 5), or a value of 17 in N(20, 4)?

- Standardize both to a z-score:
  - $z_1 = \frac{30-25}{5} = 1$ 1 sd above the mean
  - $z_2 = \frac{17-20}{4} = -0.75$ 0.75 sd below the mean

- Larger absolute value of z-score means more extreme value
- Recall from probability course!


beak_length
Counts
0
10
20
30
40
50
30
35
40
45
50
55
60
beak_length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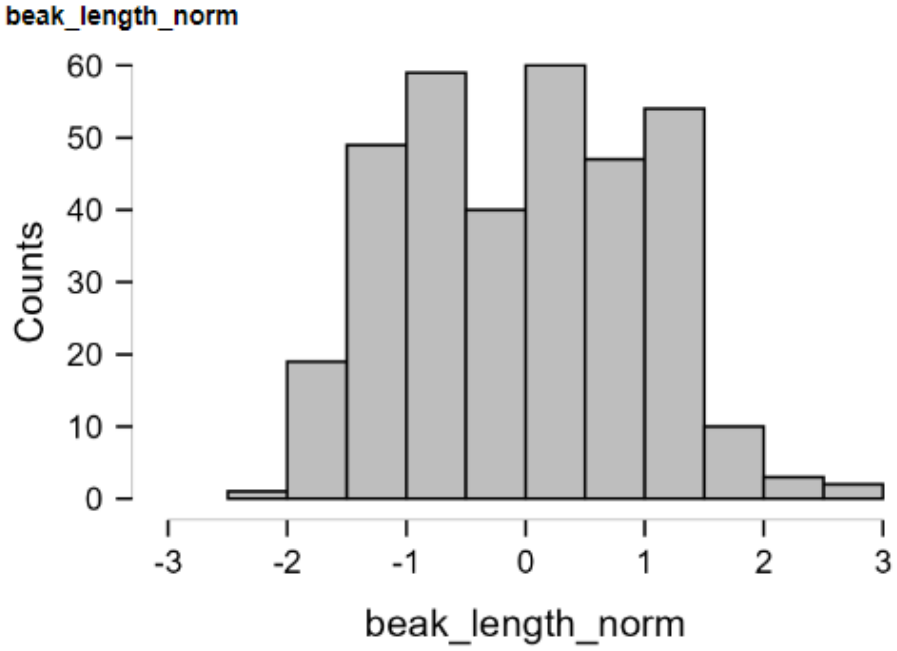
40
60
80
12
14
16
18
20
22
beak_depth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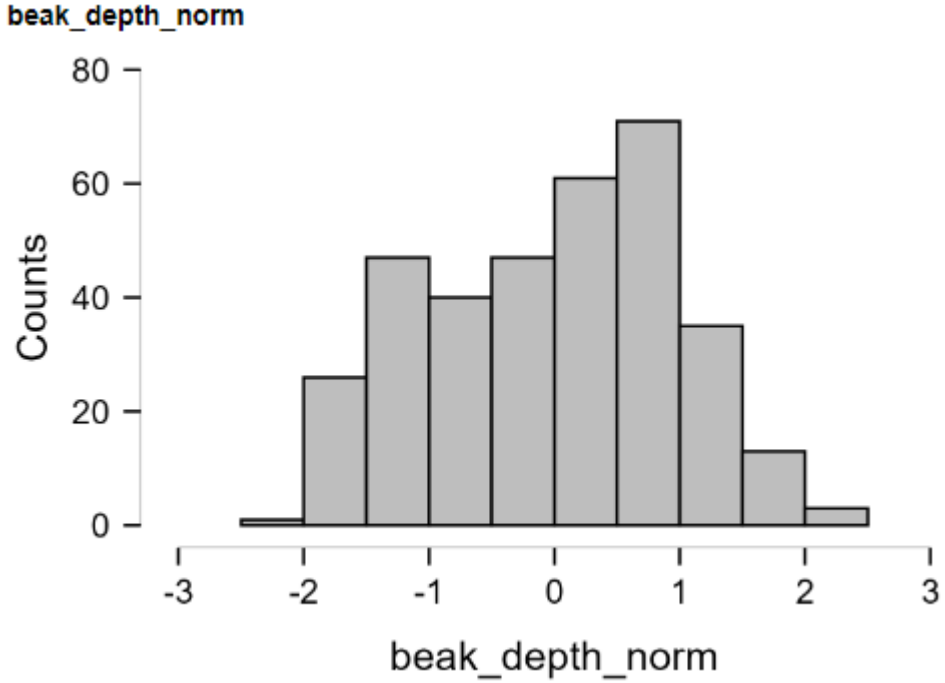
30
40
50
60
-3
-2
-1
0
1
2
3
beak_length_norm


beak_depth_norm
Counts
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20
40
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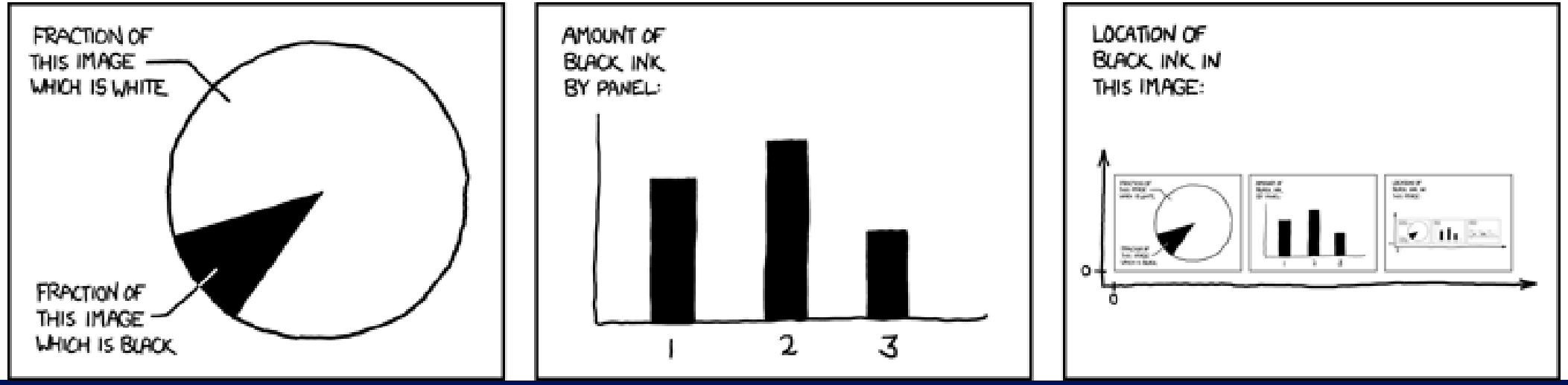
80
-3
-2
-1
0
1
2
3
beak_depth_norm

Wrapping up

- Tutorial: descriptive statistics in JASP
  - Good practice for first assignment

- Study chapter 2 in the book (also 2.7, not covered in class!)

xkcd 688